

AOS-CX Virtual Switching Framework (VSF) Guide for 6200, 6300 Switch Series



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This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing Aruba switches on a network.

Applicable products

This document applies to the following products:

- Aruba 6100 Switch Series (JL675A, JL676A, JL677A, JL678A, JL679A)
- Aruba 6200 Switch Series (JL724A, JL725A, JL726A, JL727A, JL728A)
- Aruba 6300 Switch Series (JL658A, JL659A, JL660A, JL661A, JL662A, JL663A, JL664A, JL665A, JL666A, JL667A, JL668A, JL762A)
- Aruba 6400 Switch Series (JL741A, R0X26A, R0X27A, R0X29A, R0X30A)
- Aruba 8320 Switch Series (JL479A, JL579A, JL581A)
- Aruba 8325 Switch Series (JL624A, JL625A, JL626A, JL627A)
- Aruba 8360 Switch Series (JL700A, JL701A, JL702A, JL703A, JL706A, JL707A, JL708A, JL709A, JL710A, JL711A)
- Aruba 8400 Switch Series (JL375A, JL376A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in [Support and other resources](#).

Command syntax notation conventions

Convention	Usage
<code>example-text</code>	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([]).
example-text	In code and screen examples, indicates text entered by a user.

Convention	Usage
Any of the following: ▪ <code><example-text></code> ▪ <code><example-text></code> ▪ <code>example-text</code> ▪ <code>example-text</code>	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: ▪ For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value. ▪ For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed items is required.
[]	Brackets. Indicates that the enclosed item or items are optional.
... or ...	Ellipsis: ▪ In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. ▪ In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term `switch`, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch(CONTEXT-NAME)#
```

Indicates the configuration context for a feature. For example:

```
switch(config-if)#
```

Identifies the `interface` context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100) #
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>) #
```

Where *<VLAN-ID>* is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format:

```
member/slot/port
```

On the 6100 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface 1/1/4 in software is associated with physical port 4 on the switch.

On the 6200 Switch Series

- *member*: Member number of the switch in a Virtual Switching Framework (VSF) stack. Range: 1 to 8. The primary switch is always member 1. If the switch is not a member of a VSF stack, then member is 1.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface 1/1/4 in software is associated with physical port 4 in slot 1 on member 1.

On the 6300 Switch Series

- *member*: Member number of the switch in a Virtual Switching Framework (VSF) stack. Range: 1 to 10. The primary switch is always member 1. If the switch is not a member of a VSF stack, then member is 1.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface 1/1/4 in software is associated with physical port 4 on member 1.

On the 6400 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Specifies physical location of a module in the switch chassis.
 - Management modules are on the front of the switch in slots 1/1 and 1/2.
 - Line modules are on the front of the switch starting in slot 1/3.
- *port*: Physical number of a port on a line module.

For example, the logical interface 1/3/4 in software is associated with physical port 4 in slot 3 on member 1.

On the 83xx Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface 1/1/4 in software is associated with physical port 4 on the switch.



If using breakout cables, the port designation changes to x:y, where x is the physical port and y is the lane when split to 4 x 10G or 4 x 25G. For example, the logical interface 1/1/4:2 in software is associated with lane 2 on physical port 4 in slot 1 on member 1.

On the 8400 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Specifies physical location of a module in the switch chassis.
 - Management modules are on the front of the switch in slots 1/5 and 1/6.
 - Line modules are on the front of the switch in slots 1/1 through 1/4, and 1/7 through 1/10.
- *port*: Physical number of a port on a line module

For example, the logical interface 1/1/4 in software is associated with physical port 4 in slot 1 on member 1.

Terminology

Table 1: *Acronyms used in this book*

Term	Definition
VSF	Virtual Switching Framework
L2	Layer 2 of the OSI 7-layer model
L3	Layer 3 of the OSI 7-layer model
SKU	Stock Keeping Unit
FRU	Field Replaceable Unit
ASIC	Application-Specific-Integrated Circuit
L-Agg	Link Aggregation
CLI	Command Line Interface

Table 2: *Role types*

Role	Definition
Primary	The primary member is member number 1.
Secondary	The configured secondary member.
Master	Controls VSF administration and control plane.
Standby	Standby management and under control of the master; synchronizes control plan with the master.
Member	All devices in the stack other than the master and standby are called member switches. The member switch does not run any networking protocols and has no states. The interfaces on this switch are directly controlled and programmed by the master switch.

Feature description

Virtual Switching Framework, or VSF, defines a virtual switch, comprising multiple individual physical switches, interconnected through standard Ethernet links. These physical switches will operate with one

control plane, visible to peers as a virtual switch stack. This composition simplifies management and provides the capability to scale the stack.

On-demand scalability in the access layer allows the user to increase the number of ports on a stack as per needs, without having to manage a new switch. The same stack can scale up or down to match the requirements.

- **6200F:** VSF allows stacks to be formed using any combination of SKUs of the 6200 family. Up to 8 member switches will be allowed. Connections between the switches must use 10G links.
- **6300:** VSF allows stacks to be formed using any combination of SKUs of the 6300 family. Up to 10 member switches will be allowed. Connections between the switches must use 10G, 25G, or 50G links. All VSF links in a stack should operate at the same speed.

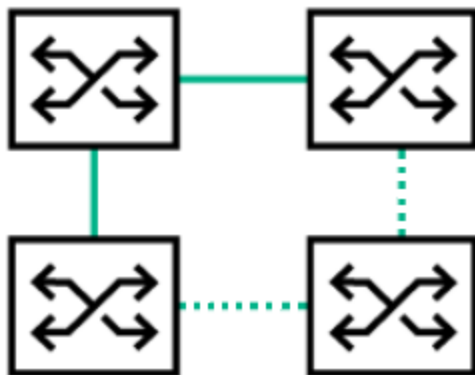
Connection topology

VSF supports up to 8 member stacks (for 6200F devices) or 10 member stacks (for 6300 devices) in ring and chain topology.

Figure 1 Chain topology



Figure 2 Ring topology



Ring is the recommended deployment topology. It inherently provides resiliency against a single failure of a link or switch.

VSF is always enabled on supported switches. Within the stack, one switch is the Master that runs all control plane software and manages the ASICs of all stack members. Any switch apart from primary can be configured as Standby switch.

VSF behavior

Each stack member must have a unique member number. When deploying a stack, ensure that each member has a distinct number by renumbering the switches to the appropriate member numbers. Stack formation will fail if there is a member number conflict.

- The primary member will become Master and the secondary member will become Standby under normal circumstances.
- The primary member is member number 1. This setting is not configurable and 1 is the default. A factory-default switch boots up as a VSF-enabled switch with a member number of 1.
- The secondary member number is user configurable, and there is no default secondary member. It is recommended that the customer configures a secondary member in the stack, since a stack with a standby offers resiliency and high-availability.
- No members other than primary and secondary members can become Master or Standby of the stack.

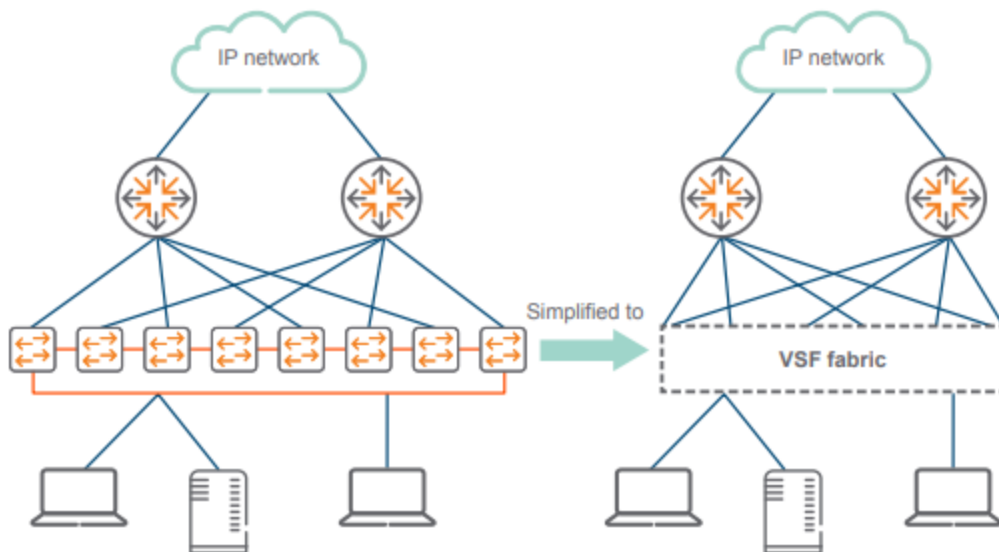


In a standard deployment, uplinks should be from primary and secondary. The management interface from primary and secondary members should be connected to the management network, providing management connectivity to the current master.

One virtual device

Once the VSF stack is formed, all interconnected switches operate as a single virtual switch with a single control plane. All interfaces of all switches in the stack are available for configuration and management.

Figure 3 One virtual device example topology



```
switch# show vsf
MAC Address           : 08:97:34:b0:0e:00
Secondary             : 2
Topology              : Ring
Status                : No Split
Split Detection Method : None

Mbr MAC Address      Type      Status
ID
-----
1  08:97:34:b0:0e:00  JL666A  Master
2  08:97:34:b1:43:00  JL665A  Standby
3  08:97:34:b7:cc:00  JL663A  Member
4  08:97:34:b6:42:00  JL662A  Member
```

Interfaces will be numbered as noted in the following table.

Name	Member Number	Slot	Port
1/1/1	1	1	1
2/1/14	2	1	14
8/1/12	8	1	12

Slot number is always fixed as 1. All interfaces are available for configuration.

```
switch# show interfaces brief
-----
Port      Native Mode  Type      Enabled Status Reason      Speed
          VLAN                                     (Mb/s)
-----
```

1/1/1	10	access	SFP+DA3	yes	up		10000
1/1/2	--	routed	--	no	down	No XCVR installed	--
1/1/3	--	routed	--	no	down	No XCVR installed	--
1/1/4	--	routed	--	no	down	No XCVR installed	--
1/1/5	--	routed	--	no	down	No XCVR installed	--
1/1/6	--	routed	--	no	down	No XCVR installed	--
...							
1/1/33	--	routed	--	no	down	No XCVR installed	--
1/1/34	--	routed	--	no	down	No XCVR installed	--
1/1/35	--	routed	--	no	down	No XCVR installed	--
1/1/36	--	routed	--	no	down	No XCVR installed	--
2/1/1	10	access	SFP+DA3	yes	up		10000
2/1/2	--	routed	--	no	down	No XCVR installed	--
2/1/3	--	routed	--	no	down	No XCVR installed	--
2/1/4	--	routed	--	no	down	No XCVR installed	--
...							
2/1/35	--	routed	--	no	down	No XCVR installed	--
2/1/36	--	routed	--	no	down	No XCVR installed	--

A single control plane operates for the entire VSF stack.

```

switch# show run
Current configuration:
!
!Version ArubaOS-CX FL.10.04.0001AQ
!export-password: default
cli-session
    timeout 0
!
!
!
!
ssh server vrf default
ssh server vrf mgmt
!
!
!
!
!
vlan 1
spanning-tree
interface mgmt
    no shutdown
    ip dhcp
interface 1/1/1
    no shutdown
    no routing
    vlan access 1
interface 1/1/2
    no shutdown
    no routing
    vlan access 1
interface 1/1/3
    no shutdown
    no routing
    vlan access 1
interface 1/1/4
    no shutdown
    no routing
    vlan access 1
interface 1/1/5

```

```

        no shutdown
        no routing
        vlan access 1
interface 1/1/6
        no shutdown
        no routing
        vlan access
...
...
interface 2/1/1
        no shutdown
        no routing
        vlan access
interface 2/1/2
        no shutdown
        no routing
        vlan access 1
interface 2/1/3
        no shutdown
        no routing
        vlan access 1
interface 2/1/4
        no shutdown
        no routing
        vlan access 1
interface 2/1/5
        no shutdown
        no routing
        vlan access
...
...
vsf secondary-member 2
vsf member 1
        type j1668a
        link 1 1/1/27
        link 2 1/1/28
vsf member 2
        type j1668a
        link 1 2/1/27
        link 2 2/1/28

```

As shown in this configuration, interfaces of all member switches can be configured from the Master. Once a stack is deployed, the stack configuration is sticky. The user can safely remove all other configurations with the command `erase startup-configuration` without disturbing the stack configurations. To remove all configurations, including the stacking configurations, use the command `erase all zeroize`, where all members of the stack will be reset to factory defaults

Interoperation

The VSF stack supports either:

- 6200F devices, or
- 6300 devices (6300M or 6300F).



VSF stacking cannot be done with a mixed set of switches. The stack must be made up of *only* 6200 or *only* 6300 switches.



Link aggregation

Link aggregations (L-Agg) can span interfaces across multiple individual switches within the stack. Load balancing is performed on all interfaces of the L-Agg across the stack and is applicable to L2 and L3 L-Aggs.

```
interface lag 1
    no shutdown
    no routing
    vlan access 1
    loop-protect
interface lag 2
    no shutdown
    bfd min-transmit-interval 1000
    ip address 192.168.12.7/24
interface 1/1/18
    no shutdown
    lag 1
interface 2/1/18
    no shutdown
    lag 1
interface 1/1/23
    no shutdown
    lag 2
interface 2/1/23
    no shutdown
    lag 2
```

```
switch# show lacp interfaces
```

State abbreviations :

A - Active P - Passive F - Aggregable I - Individual
S - Short-timeout L - Long-timeout N - InSync O - OutofSync
C - Collecting D - Distributing
X - State m/c expired E - Default neighbor state

Actor details of all interfaces:

Intf	Aggr Name	Port Id	Port Pri	State	System-ID	System Pri	Aggr Key	Forwarding State
1/1/18	lag1							up
2/1/18	lag1							up
1/1/23	lag2							up
2/1/23	lag2							up

Partner details of all interfaces:

Intf	Aggr Name	Port Id	Port Pri	State	System-ID	System Pri	Aggr Key
1/1/18	lag1						
2/1/18	lag1						

```
1/1/23 lag2  
2/1/23 lag2
```

Consoles

The serial console of the Master switch provides a full CLI configuration interface for a user with valid credentials. The serial console of the other stack members, including the Standby, provides a reduced CLI configuration interface, with only a limited set of commands for troubleshooting the stack.

In a standard deployment, connect to the console interface of the master and standby switch. This enables the stack master console to be reachable after a stack failover to the new Master.



Any switch configuration or monitoring must be performed from the console of the stack Master switch only.

Management interface

In a VSF stack, only the management interface on the Master switch will be assigned an IP address (configured or assigned by DHCP). The stack allows connectivity to management protocols and Console through the management interface on the Master.

VSF configuration

The following aspects of VSF are user-configurable.

Member number

To add a device to a VSF stack, the device must be renumbered to the corresponding member ID. The user can specify the member number of the switch. The default member number is 1.

- For the 6200F device, the default number can be changed to any value from 2 through 8. (The device supports up to 8 members.)
- For 6300 devices, the default number can be changed to any value from 2 through 10. (The device supports up to 10 members.)

Refer to [vsf renumber-to](#) and [Misconfiguration recovery](#) for information about renumbering a member.



Changing the member number causes the switch to reboot and all configuration on the switch is removed.

A switch with a member number other than 1 cannot boot completely unless it has reachability to a VSF master switch via VSF link. If a renumbered member is unable to communicate with the master switch and is waiting in booting state, the user can:

- Go to a recovery console with a `ctrl+c` sequence and collect the diagnostic information, or
- Reset the VSF configuration.

Access to VSF members

In addition to serial console connections, any stack member can be accessed from any other member using the `member` command.

Refer to [member](#) for information about console connection to a member switch.

VSF links

The user can specify the interfaces which comprise the VSF links. Refer to [link](#) for information about specifying interfaces.

When the interface is configured, any existing configuration is removed, including VLAN memberships, ACL/Quality of Service rules and any speed/duplex/MTU configuration.

Once the interface becomes part of a VSF link, no protocol or feature will be allowed to run on it as it is now part of the fabric.



A VSF link will be a routed interface.

Member provisioning

VSF allows the user to provision or pre-configure any member before the member is physically added to the stack. Provisioning the member allows the user to complete the required configuration as if the member is present in the stack. When the member eventually joins the stack, it will boot up with the configuration made on the pre-provisioned interfaces.

To provision a member, the part number of the member must be specified. Refer to [type](#) for information about provisioning a member.



If a member tries to join the stack with a different part number to the one provisioned on the Master, the member will be removed from the stack and will reboot with factory defaults.

Secondary member

The stack will not have a standby member by default. A secondary member can be configured from available members and it will be assigned the role of stack standby.



Member number 1 can never be configured as a secondary member.

When configured as secondary, a stack member that is already present in the stack will reboot and rejoin the stack as the standby.

A provisioned member can be configured as a secondary member. When the member joins the stack, it will boot up in the standby role, without any further reboot.

If a secondary member is already configured and physically present in the stack, removing the secondary will cause the secondary member to reboot and join as a member.

Refer to [vsf secondary-member](#) for information about configuring a secondary member.

To change secondary configuration, first unconfigure the secondary, wait for the device to join as a member, and then configure a new secondary

Member remove

A member can be removed from a running stack. All configuration associated with the member will be removed.

If the member is physically present in the stack at the time it is removed, all VSF configurations on that member will be erased and it will lose its identity as a member of the stack from which it was removed. The member will come back as member 1 with factory default configuration.



It is not advisable to remove the member that is the master of the stack. If the master has to be removed, the recommendation is to switch over and wait for the old master to come up as standby before removing it.

Refer to [vsf member](#) for information about removing a member.

Automated image sync

In a VSF environment, all stack members run the same software image. If the user upgrades the software on the Master by downloading a new software image using SFTP/TFTP, all members of the stack will simultaneously upgrade.

When forming a stack, if the software version on a member is different from the version of the Master, the member will automatically update itself to the same version as the Master. The member will reboot itself to run the updated version before joining the stack.

Reboot

An individual stack member can be rebooted from a CLI command.

- The member will reboot and re-join the stack, with the same role that it had prior to the reboot.
- If the stack topology is a ring, no traffic disruption is expected on any other stack members when a single member is rebooted.
- If the stack topology is a chain, rebooting a member may cause a stack split, resulting in members being unreachable from the master. This result can cause significant disruption of the stack, so use this option with caution.
- If the member is the stack Standby, there will be no Standby in the stack until the member reboots and re-joins the stack. At this point, the member will again have the role of Standby.
- If the member is the stack Master, the command will trigger a failover and the Standby switch will take over as Master of the stack.
- If the Standby is unavailable at the time of master reboot, the whole stack will reboot.

The whole stack can also be rebooted by using the `boot system` command.

- All members will reboot and the stack will re-form.
- Traffic will be disrupted for the duration of the reboot.

Refer to [vsf member reboot](#) for information about rebooting a member.

Member addition

A member can be added to the stack to augment an existing stack. The member being added can be a factory-default switch or a switch with pre-existing configuration.

1. Configure interfaces to VSF links on the member being added.
2. Renumber the member being added.



The member will not join the stack if there is a member number conflict.

3. Renumbering will cause a reboot of the switch.
4. Connect the configured VSF links to a previously configured VSF link on the stack.
5. The member joins the stack, with default configuration on its interfaces. Any previous configuration on the member will be lost.

Member replacement

The replacement member must be of the same part number as the switch being replaced.

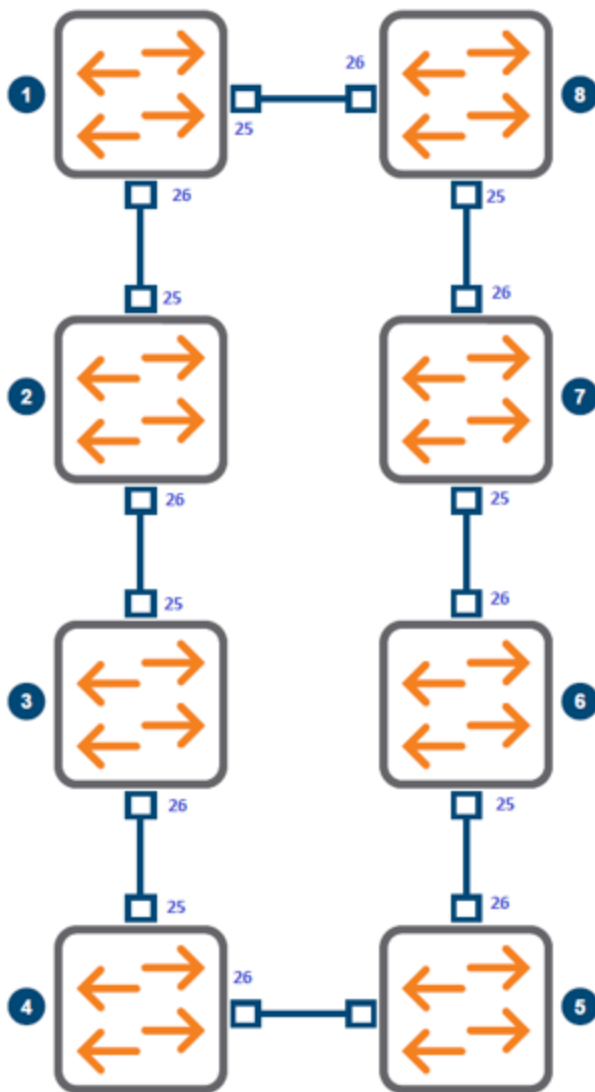
1. Power off or disconnect all physical connections of the member that will be replaced.
2. Configure interfaces to VSF links on the replacement member. These interfaces **must** match the interfaces configured on the switch being replaced.
3. Renumber the replacement member to the same number as the switch being replaced.
4. Renumbering will cause a reboot of the switch.
5. Connect the replacement member to the stack.
6. The member joins the stack, with the same configuration as the member it is replacing.

The following sections describe the prerequisites and procedures to configure a VSF stack.

Prerequisites

Manual configuration of a VSF stack requires the user to individually configure each switch in the stack. This process provides the best control for the user to configure VSF member number and links.

Figure 4 *Eight-member ring setup*



Forming an eight-member ring setup

To form an eight-member ring setup as shown, do not make the connections initially. Connect the ports only after each device is fully configured.

Procedure

1. Log in to the first device, numbered 1.
 - a. The default member number is 1, so no member number change is required.
 - b. At the prompt, enter the following commands:

```
switch# configure
switch(config)# vsf member 1
switch(vsf-member-1)# link 1 1/1/25
switch(vsf-member-1)# link 2 1/1/26
```

- c. The preceding sequence of commands will configure the links for member 1.
 - d. Ports 25 and 26 are configured as link 1 and 2 respectively.
2. Log in to the second device, numbered 2.
 - a. Execute the following commands.

```
switch# configure
switch (config)# vsf member 1
switch(vsf-member-1)# link 1 1/1/25
switch(vsf-member-1)# link 2 1/1/26
switch(vsf-member-1)# exit
switch(config)# vsf renumber-to 2
This will save the VSF configuration and reboot the switch.
Do you want to continue (y/n)? y
```

- b. The preceding sequence of commands will configure the links on member 2.
 - c. The default member number is "1". The command "vsf renumber-to" changes this member number.
 - d. Links are configured **before** renumbering, and the member identifier in the interface name is "1" at this point.
 - e. The switch will reboot after executing the renumber command.
3. Physically connect member 2 to member 1 as shown in the figure.
 - a. This action will cause member 2 to join the stack, with member 1 as the master.
 - b. This result can be verified by executing "show vsf" on member 1.

```
switch# show vsf
MAC Address           : 38:21:c7:5d:d0:c0
Secondary             :
Topology              : Chain
Status                : No Split
Split Detection Method : None

Mbr Mac Address      type      Status
ID
---
1  38:21:c7:5d:d0:c0  JL668A  Master
2  38:21:c7:6a:10:c0  JL668A  Member
```

4. Repeat steps 2 and 3, for each stack member 3 through 8.
 - a. Be sure to specify the member number correctly on each member.
 - b. If a member number conflict is detected, the member will NOT join the stack.
5. Once member 8 has successfully joined the stack, connect member 8 link 2 to member 1 link 1, to complete the ring.

Issue a "show vsf" command to ensure that the ring has successfully formed.

```
switch# show vsf
MAC Address      : 38:21:c7:5d:d0:c0
Secondary       :
Topology        : Ring
Status          : No Split
Split Detection Method : None

Mbr Mac Address      type      Status
ID
---
1  38:21:c7:5d:d0:c0  JL668A  Master
2  38:21:c7:6a:10:c0  JL668A  Member
3  38:21:c7:5c:15:80  JL668A  Member
4  38:21:c7:5a:61:40  JL668A  Member
5  38:21:c7:62:66:00  JL668A  Member
6  38:21:c7:58:22:40  JL668A  Member
7  38:21:c7:5a:9c:00  JL668A  Member
8  38:21:c7:63:a5:00  JL668A  Member
```

6. The preceding steps will form an eight-member stack without a standby. To make any member the standby (for example, member 8), use the secondary command:
 - a. From the primary VSF member, configure member 8 as VSF secondary member:

```
switch(config)# vsf secondary-member 8
This will save the configuration and reboot the specified switch.
Do you want to continue (y/n)? y

switch(config)#
```

- b. This action will reboot member 8 and it will rejoin as standby.

```
switch# show vsf
MAC Address : 38:21:c7:5d:d0:c0
Secondary : 8
Topology : Ring
Status : No Split
Split Detection Method : None
Mbr Mac Address type Status
ID
---
1  38:21:c7:5d:d0:c0  JL668A  Master
2  38:21:c7:6a:10:c0  JL668A  Member
3  38:21:c7:5c:15:80  JL668A  Member
4  38:21:c7:5a:61:40  JL668A  Member
5  38:21:c7:62:66:00  JL668A  Member
6  38:21:c7:58:22:40  JL668A  Member
```

```
7 38:21:c7:5a:9c:00 JL668A Member
8 38:21:c7:63:a5:00 JL668A Standby
```

7. Alternatively, before adding member 8 to the stack, pre-configure the secondary as 8 and then renumber device 8. This action will ensure that device 8 will join the stack directly as standby.

Misconfiguration recovery

If a switch fails to join the stack because of misconfiguration, use the following procedure to restore the switch back to a factory default configuration.



The user must have management connectivity to the failed member for support files from the member in recovery mode.

Procedure

1. Press **Ctrl+C** on the switch console.
2. Log in using administrator credentials.
3. At the prompt, issue the `vsf-factory-reset` command.

```
^C
Login: admin
Password:

recovery# vsf-factory-reset
```

4. This resets the member to factory-default settings and the switch will come up with a default member ID of 1.
5. Now the user can reconfigure the VSF link and renumber it to the preferred member ID.

The following section describes failure and recovery scenarios for stack split issues.

Stack split

Multiple link or member failures can result in a complete stack split, with the master and standby switch in different split fragments. In this case, both master and standby switches will become master switches. Each stack fragment will continue operating with the same configuration and state, and forward traffic between all stations that the fragment can reach.

The downside of this scenario is that each stack fragment will have the same MAC address and IP addresses. To avoid this scenario, configure split detection (described in the following section) which would bring down the interfaces on one fragment to prevent duplicate MAC/IPs.

Management interface split detection



For more information, refer to [vsf split-detect](#).

VSF stack supports management split detection, which requires users to connect the management interfaces of the primary and secondary stack members to the same L2 network.

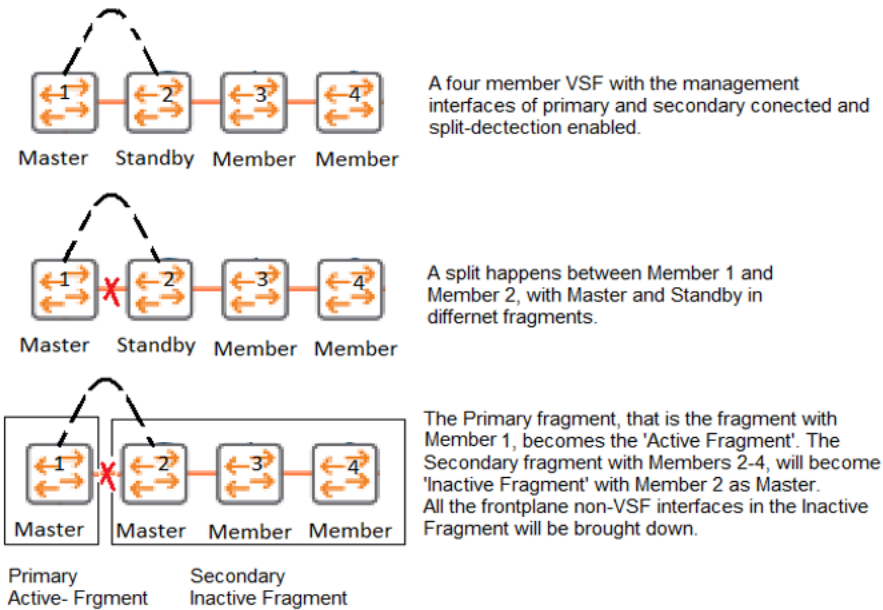
The Primary stack member is member "1", whereas the secondary member is the user-configured secondary switch. Once the stack is split, both of these switches become master of their respective fragments. The fragment with "1" as master is referred to as the primary fragment, and the fragment with the secondary switch as master is the secondary fragment.



It is also possible to connect the management interfaces of primary and secondary to one another for split detection.

If the secondary fragment discovers that the primary fragment is operational, it will bring down all front-plane non-VSF interfaces on the secondary fragment to minimize network disruption due to duplicate MAC or IP addresses.

The interfaces will remain down until the stack is reconnected or the primary fragment goes down. The interfaces of the primary fragment will always remain operational.



The `show vsf` output in the Primary fragment will look like this:

```
switch# show vsf

MAC Address           : 38:21:c7:5c:f4:c0
Secondary             : 2
Topology              : Standalone
Status                : Active Fragment
Split Detection Method : mgmt

Mbr Mac Address      type      Status
ID
-----
1  38:21:c7:5c:f4:c0  JL668A  Master
2                      JL668A  In Other Fragment
3                      JL668A  In Other Fragment
4                      JL668A  In Other Fragment
switch#
switch# show vsf topology
Mstr
+---+
| 1 |
+---+
switch#
```

The `show vsf` output in the secondary fragment will look like this:

```
switch# show vsf

MAC Address           : 38:21:c7:5c:f4:c0
Secondary             : 2
Topology              : Chain
Status                : Inactive Fragment
Split Detection Method : mgmt

Mbr Mac Address      type      Status
```

```

ID
-----
1          JL668A      In Other Fragment
2  38:21:c7:5c:77:40  JL668A      Master
3  38:21:c7:5a:a5:80  JL668A      Member
4  38:21:c7:5c:b3:00  JL668A      Member
switch#
switch# show vsf topology
          Mstr
+---+   +---+   +---+
| 4 |1==2| 3 |1==2| 2 |
+---+   +---+   +---+
switch#

```

The following recommendations and restrictions apply to VSF.

- Before applying a configuration on a stack through checkpoint restore or TFTP/SFTP/USB download, make sure that current VSF-specific configurations and the intended configurations match exactly. In other words, the VSF stack and the intended configuration must have the same:
 - Total number of members
 - Member types
 - Member number/ID
 - VSF link configurations
 - Secondary member configuration
 - Split-detect configuration
- A functional stack must be configured with a standby for redundancy purposes. If the master fails and there is no standby, the stack will fail.
- If the master fails and there is a standby device, the standby becomes the new master and will take over stack management. When the old master device is replaced, it seamlessly becomes the standby device for the stack and there no disruption.

The MAC address of the stack will remain the same until the entire stack is rebooted, after which the stack MAC address will be the MAC address of the new master. However, once recovered, it is not advisable to use the removed master elsewhere in the same network until the stack is rebooted to avoid MAC address conflicts.

- After downloading firmware to a stack, the stack must be rebooted to complete the upgrade process. Adding or rebooting individual members before the upgrade process is completed can cause the individual member to fail while joining the stack.
- If there is a discrepancy between a VSF member link configuration on the master and the VSF member link configuration on the member, the link configuration on the member is used.
- Configure only one link between a pair of stack members. This recommendation is also applicable for a two-member VSF stack.
- If there is a split, failure in the connectivity between management interfaces of the master and standby might result in two active fragments. This issue can occur even if management split-detect is enabled.
- Ring topology is not supported for two-member VSF.
- Replacing member 1 in a stack without a standby with a new switch booted as member 1 will reset all configurations on the stack.
- Do not connect a renumbered member to multiple primary devices through VSF links.
- Before removing an individual interface from VSF link using the command `no vsf link <x> <interface>`, ensure that the interface is admin shutdown at both local and peer ends. For example: Interface 1/1/25 on member 1 link 1 is connected to 2/1/25 on member 2 link 2. The user intends to remove 1/1/25 from link 1 of member 1. Both the interfaces 1/1/25 and 2/1/25 have to be admin shutdown before actually removing them from the link configuration. To delete the link completely

using the `no link <x>` command, all individual interfaces in the VSF link have to be admin shutdown both at local and peer ends.

- Before removing an individual interface from the VSF link using the `no vsf link <x> <interface>` command, ensure that the interface is admin shutdown at both local and peer ends.

Example: Interface 1/1/25 on member 1 link 1 is connected to 2/1/25 on member 2 link 2. The user intends to remove 1/1/25 from link 1 of member 1. Interfaces 1/1/25 and 2/1/25 must be admin shutdown before removing them from the link configuration.

Before deleting the link using the `no link <x>` command, all individual interfaces in the VSF link must be admin shutdown both at local and peer ends.

- There may be instances in which a master switch with `vsf secondary <id>` configuration is unable to discover the standby switch. In such cases, the master switch will wait for up to 10 minutes to detect the standby switch.

VSF commands do not apply to the 6400 series switches.

vsf member

Syntax

```
vsf member <MEMBER-ID>  
no vsf member <MEMBER-ID>
```

Description

Creates VSF member context in the switch for the specified member.

The `no` form of this command removes the specified member from the stack. All configuration associated with the member, as well as the subsystems and interfaces of the member will also be removed.

If the member is physically present in the stack at the time it is removed, it will reboot with the default configuration and lose its identity as a member of the stack from which it was removed.



When a physically present member is removed, it may cause the stack to split.

Command context

config

Parameters

<MEMBER-ID>

VSF member identifier. Required.

- Range for 6200F devices: 1-8.
- Range for 6300 devices: 1-10.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring a VSF member:

```
switch(config)# vsf member 2  
switch(vsf-member-2)#
```

Removing a non-master member from the stack:

```
switch(config)# no vsf member 2
The specified switch will be unconfigured and rebooted
Do you want to continue (y/n)? y
```



Removing the running master should be done with caution as it can make the stack unusable if there is no standby.

member

Syntax

```
member <MEMBER-ID>
```

Description

Connects to the specified member in a VSF environment.

Command context

Manager (#)

Parameters

<MEMBER-ID>

VSF member ID. Required.

- Range for 6200F devices: 1-8.
- Range for 6300 devices: 1-10.

Authority

Administrators or local user group members with execution rights for this command.

Examples

VSF stack is formed with two members:

```
switch# member 2
admin@172.17.17.2's password:

Last login: 2019-09-30 11:42:17 from the console
User "admin" has logged in 1 time in the past 30 days
member#
```

Member to self:

```
switch# member 1
Already on member id 1
```

VSF stack is not formed and member not available:

```
switch# member 2
No stack role for member id 2
```

type

Syntax

```
type <TYPE>
```

Description

Configures the part number of the VSF member being provisioned. After provisioning, the interfaces of the member are available for configuration.

When the member eventually joins the stack, it will boot up with the configuration made on the pre-provisioned interfaces.

To provision a member, the member number and the part number of the member must be specified.

Command context

```
vsf-member-<ID>#
```

Parameters

<TYPE>

The part number of the member being provisioned. Required.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the part number of a VSF member:

```
switch(vsf-member-2)#
  type The part number of the member being provisioned

switch(vsf-member-2)# type ?
  j1658a  6300M 24SFP+ /4SFP56 Switch
  j1659a  6300M 48SR PoE CLS 6 /4SFP56 Switch
  j1660a  6300M 24SR PoE CLS 6 /4SFP56 Switch
  j1661a  6300M 48G PoE CLS 4 /4SFP56 Switch
  j1662a  6300M 24G PoE CLS 4 /4SFP56 Switch
  j1663a  6300M 48G /4SFP56 Switch
  j1664a  6300M 24G /4SFP56 Switch
  j1665a  6300F 48G PoE CLS 4 /4SFP56 Switch
  j1666a  6300F 24G PoE CLS 4 /4SFP56 Switch
  j1667a  6300F 48G /4SFP56 Switch
  j1668a  6300F 24G /4SFP56 Switch

switch(vsf-member-2)# type j1662a

switch(vsf-member-2)# show running-config
Current configuration:
!
!Version ArubaOS-CX
!
!
```

```

!
!
ssh maximum-auth-attempts 6
!
!
!
!
!
vlan 1
vsf member 1
    type j1661a
exit
vsf member 2
    type j1662a
exit

```

link

Syntax

```
link <LINK-ID> [<IFRANGE>]
```

Description

Creates or modifies a VSF link. The user can specify the physical interfaces that make up the VSF link. Once an interface is part of a VSF link, all existing configuration on the interface is removed and the interface will operate as a VSF interface. At least one interface must be specified for the creation of a VSF link. VSF interfaces carry VSF traffic and can only be connected to other VSF interfaces.

The `no` form of the command can be used to remove interfaces from a link or remove configuration from the link completely.



When configuration is removed from a link, it may cause the stack to split.

Command context

```
vsf-member-<ID>#
```

Parameters

<LINK-ID>

The VSF link number. Required. Range: 1-2.

<IFRANGE>

The interface identifier range. Required.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating and modifying VSF links:

```

switch(vsf-member-1)# link
<1-2>          VSF Link number
switch(vsf-member-1)# link 1

```

```

IFRANGE      Interface identifier range
<cr>
switch(vsf-member-1)# link 1 1/1/51
<cr>
switch(vsf-member-1)# link 1 1/1/49-1/1/50
<cr>
switch(vsf-member-1)# link 2 1/1/52
<cr>
switch(vsf-member-1)# link 1 1/1/49
<cr>

switch(vsf-member-5)# show running-config
Current configuration:
!
!Version ArubaOS-CX
!
!
!
ssh maximum-auth-attempts 6
!
!
!
!
vlan 1
interface 1/1/49
    no shutdown
interface 1/1/52
    no shutdown
vsf member 1
    type jl661a
    link 1 1/1/49
    link 2 1/1/52
exit

switch(vsf-member-1)# no link 1
<cr>
switch(vsf-member-1)# no link 1
This will cause the stack to split.
Do you want to continue (y/n)? y

switch(vsf-member-1)# no link 2
This will cause the stack to split and the residual stack fragment will become
unusable.
Do you want to continue (y/n)? y

switch(vsf-member-1)# show running-config
Current configuration:
!
!Version ArubaOS-CX SL.10.02.0020-741-g11104d6~dirty
!
!
!
!
ssh maximum-auth-attempts 6
!
!
!
!
vlan 1

```

```
interface 1/1/52
  no shutdown
vsf member 1
  type jl661a
exit
```

Before removing an individual interface from the VSF link using the `no vsf link <x> <interface>` command, ensure that the interface is admin shutdown at both local and peer ends.

Example: Interface 1/1/25 on member 1 link 1 is connected to 2/1/25 on member 2 link 2. The user intends to remove 1/1/25 from link 1 of member 1. Interfaces 1/1/25 and 2/1/25 must be admin shutdown before removing them from the link configuration.

Before deleting the link using the `no link <x>` command, all individual interfaces in the VSF link must be admin shutdown both at local and peer ends.

vsf split-detect

Syntax

```
vsf split-detect <MGMT-INTERFACE>
```

Description

Configures the VSF split detection method that specifies the mechanism used for stack fragment discovery when there is a stack split.

Once the stack fragments are discovered, the fragment having the primary member always wins. All non-VSF interfaces on the losing stack fragment will be brought down to minimize network disruption due to duplicate MAC/IP.

Command context

```
config
```

Parameters

<MGMT-INTERFACE>

Configures mgmt-interface as the split detection method. Connect the management interfaces of the primary and secondary members to the same L2 network. Optionally, the management interfaces of primary and secondary can be directly connected to each other.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring mgmt-interface as the split detection method:

```
switch(config)# vsf split-detect mgmt-interface
```

Removing split detection from the stack:

```
switch(config)# no vsf split-detect
```

vsf secondary-member

Syntax

```
vsf secondary-member <MEMBER-ID>
```

Description

Configures a secondary member from the available members. The secondary member will normally operate as the Standby member of the stack.

The stack does not have a secondary member by default. If a secondary member is already configured, the user must un-configure the old secondary member and then configure new secondary member.



Member 1 cannot be configured as the secondary member.

Command context

config

Parameters

<MEMBER-ID>

Secondary member number. Required.

- Range for 6200F devices: 2-8.
- Range for 6300 devices: 2-10.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring and un-configuring a secondary member:

```
switch(config)# vsf secondary-member 3
This will save the configuration and reboot the specified switch.
Do you want to continue (y/n)? y

switch(config)# vsf secondary-member 4
The existing secondary member must be unconfigured first.

switch(config)# no vsf secondary-member
The secondary member will go for a reboot.
Do you want to continue (y/n)? y
```

vsf renumber-to

Syntax

```
vsf renumber-to <MEMBER-ID>
```

Description

Renumbers VSF member 1 to a value from 2 through 10 (for 6300 devices) and 2 through 8 (for the 6200F device). Changing the member number causes the switch to reboot with the new member number. Only member 1 can be renumbered.



VSF links must be configured before renumbering a switch. Renumbering will be disallowed if no links are configured or there are provisioned/physically present members.

Command context

config

Parameters

<MEMBER-ID>

Member number to which the member will be renumbered. Required.

- Range for 6200F devices: 2-8.
- Range for 6300 devices: 2-10.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Renumbering primary VSF member from 1 to 2:

```
switch(config)# vsf renumber-to 2
Member 1 cannot be renumbered until all other members are removed.

switch(config)# vsf renumber-to 2
Member 1 cannot be renumbered until a VSF link is configured.

switch(config)# vsf renumber-to 2
This will save the VSF configuration and reboot the switch.
Do you want to continue (y/n)? y
```

vsf member reboot

Syntax

vsf member <MEMBER-ID> reboot

Description

Reboots the specified VSF member. Upon reboot, if the master is reachable, the member will rejoin the stack.

Command context

Manager (#)

Parameters

<MEMBER-ID>

Member number to be rebooted. Required.

- Range for 6200F devices: 1-8.
- Range for 6300 devices: 1-10.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Rebooting the primary switch of the stack:

```
switch# vsf member 1 reboot
Rebooting the master switch of the stack without a standby
will make the stack unusable.
Do you want to continue (y/n)? y

switch# vsf member 1 reboot
The master switch will reboot and the standby will become the master.
Do you want to continue (y/n)? y

switch# vsf member 2 reboot
This will reboot the specified switch.
Do you want to continue (y/n)? y
```

interface

Syntax

```
interface <IFRANGE>
```

Description

Enters configuration context for one or more VSF link interfaces.

Command context

```
config
```

Parameters

<IFRANGE>

PORT identifier range. Required.



VSF link interfaces cannot be included in a range with other interfaces.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Entering configuration context:

```
switch(config)# int 1/1/1
```

shutdown

Syntax

```
shutdown
```

Description

Shuts down one or more VSF link interfaces.

Command context

config-if-vsfc

Authority

Administrators or local user group members with execution rights for this command.

Examples

Shutting down a VSF link interface:

```
switch(config)# int 1/1/1-1/1/2
switch(config-if-vsfc-<1/1/1-1/1/2>)# shutdown
```



Shutdown configuration for VSF interfaces is not persistent across reboots.

show vsf

Syntax

show vsf

Description

Displays the list of provisioned VSF stack members.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```
switch# show vsf
MAC Address      : 08:97:34:b0:0e:00
Secondary        : 2
Topology         : Ring
Status           : No Split
Split Detection Method : None

Mbr MAC Address  Type      Status
ID
-----
1  08:97:34:b0:0e:00  JL666A   Master
2  08:97:34:b1:43:00  JL665A   Standby
3  08:97:34:b7:cc:00  JL663A   Member
4  08:97:34:b7:cc:00  JL662A   Not Present
```

```

switch# show vsf
MAC Address           : 08:97:34:b0:0e:00
Secondary             : 2
Topology              : Ring
Status                : Active Fragment
Split Detection Method : mgmt-interface

Mbr MAC Address      Type           Status
ID
-----
1  08:97:34:b0:0e:00  JL666A           Master
2  08:97:34:b1:43:00  JL665A           In Other Fragment
3  08:97:34:b7:cc:00  JL663A           Member
4                               JL662A           Not Present

```

show vsf detail

Syntax

```
show vsf detail
```

Description

Displays detailed information related to the current state of the VSF stack and the stack members.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```

switch# show vsf detail

VSF Stack
  MAC Address           : ec:eb:b8:d0:80:40
  Secondary             : 2
  Topology              : Ring
  Status                : No Split
  Uptime                : 0d 0h 23m
  Split Detection Method : None
  Software Version       : SL.10.02.0000-7755

  Name                  : Aruba-VSF-6300F
  Contact                :
  Location               :

Member ID              : 1
  MAC Address           : ec:eb:b8:d0:80:40
  Type                  : JL666A
  Model                 : Aruba 6300F 24G PoE CLS 4 /4SFP56 Switch
  Status                : Master
  ROM Version           : SL.10.02.0000-7755
  Serial Number         : CN7ZK90012
  Uptime                : 0d 0h 23m

```

```

CPU Utilization      : 0%
Memory Utilization   : 20%
VSF link 1           : Up, connected to peer member 2, link 1
VSF link 2           : Down

Member ID            : 2
MAC Address          : eb:ec:d8:e0:50:60
Type                 : JL666A
Model                : Aruba 6300F 24G PoE CLS 4 /4SFP56 Switch
Status               : Standby
ROM Version          : SL.10.02.0000-7755
Serial Number        : CN7ZK90012
Uptime               : 0d 0h 23m
CPU Utilization      : 0%
Memory Utilization   : 15%
VSF link 1           : Up, connected to peer member 1, link 1
VSF link 2           : Down

Member ID            : 3
MAC Address          :
Type                 : JL666A
Model                : Aruba 6300F 24G PoE CLS 4 /4SFP56 Switch
Status               : Not Present
ROM Version          :
Serial Number        :
Uptime               :
CPU Utilization      :
Memory Utilization   :
VSF link 1           :
VSF link 2           :

```

show vsf link

Syntax

```
show vsf link
```

Description

Displays the VSF link state for each member.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```
switch# show vsf link
```

```
VSF Member 1
```

Link	Link State	Peer Member	Peer Link	Interfaces
----	-----	-----	----	-----

1	Down	0	0	1/1/1,1/1/5,1/1/8,1/1/9
2	Down	0	0	1/1/11-1/1/13,1/1/15

show vsf member

Syntax

```
show vsf member <MEMBER-ID>
```

Description

Displays information about the specified VSF member.

Command context

Manager (#)

Parameters

<MEMBER-ID>

VSF member identifier. Required.

- Range for 6200F devices: 1-8.
- Range for 6300 devices: 1-10.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```
switch# show vsf member 1

Member ID           : 1
MAC Address         : ec:eb:b8:d0:80:40
Type                : JL557A
Model               : Aruba JL557A 2930F-48G-740W-PoE+-4SFP Switch
Status              : Master
ROM Version         : SL.10.02.0000-7755
Serial Number       : CN7ZK90012
Uptime              : 0d 0h 18m
CPU Utilization     : 0%
Memory Utilization  : 15%
VSF link 1          : Down
VSF link 2          : Down
```

show vsf topology

Syntax

```
show vsf topology
```

Description

Displays information about VSF stack member connections.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

```
switch# show vsf topology
```

	Stby	Master
+---+	+---+	+---+
3 1==2	2 1==1	1
+---+	+---+	+---+

What is VSF?

Virtual Switching Framework, or VSF, defines a single virtual switch comprised of multiple individual physical switches that are interconnected through standard Ethernet links. These links are referred to as VSF links.

These physical switches will function as one device with a unified control and management plane. Multiport VSF links are supported.

What are the supported platforms for VSF?

The Aruba 6300F/M Switch Series supports VSF.

VSF can be formed with a combination of any of the Aruba 6300F/M Switch Series (JL658A, JL659A, JL660A, JL661A, JL662A, JL663A, JL664A, JL665A, JL666A, JL667A, JL668A) or a combination of any of the Aruba 6200F Switch Series (JL724A, JL725A, JL726A, JL727A, JL728A).



Aruba 6200F Switch Series only supports fixed SKUs.

What port speeds do VSF links support?

For Aruba 6300F/M Switch Series: All uplink ports with 10G, 25G, and 50G speeds can be configured as VSF links.

For Aruba 6200F Switch Series: All uplink ports with 10G speed can be configured as VSF links.

Aruba recommends that all VSF links be configured to run at the same speed.

Can VSF be disabled?

Users cannot disable VSF. A factory default switch boots up as a VSF-enabled device with its Member ID set to 1.

What is a primary switch in VSF stack? Is it configurable?

Only the switch with a Member ID of 1 will be the primary switch in a VSF stack. This switch will function as the stack master and will drive the control and management plane for the stack.

What is a secondary switch in a VSF stack? Is it configurable?

The secondary switch will function as the standby in a stack. There is no secondary switch by default. Any member other than Member 1 can be configured as the secondary switch using the `vsf secondary-member <MEMBER-ID>` command.

Aruba strongly recommends that you configure a secondary member (standby) for stack high-availability.

How many secondary member switches are configurable in a VSF stack?

A VSF stack can be configured with one secondary member only.

Once it is configured, is it possible to change the secondary member?

Yes. Remove the current secondary member using the `no vsf secondary-member` command. This action will trigger the member to reboot and join the stack (not a standby anymore).

A new secondary member can be configured using the `vsf secondary-member <MEMBER-ID>` command. The device will reboot and rejoin the stack back as standby.

The secondary member configuration can only be changed when Member 1 is master of the stack.

How are master and standby for a stack determined?

By default, the primary member (Member 1) becomes the master of the stack and the user-configured secondary member becomes the standby.

The secondary member synchronizes all its states with the master. If the current master (Member 1) fails, the standby (secondary member) will seamlessly transition to the master role. In this state, if Member 1 comes back up, it will take the standby role.

Only primary and secondary members can take up master and standby roles in a stack.

What is the role of other members in a stack?

All devices other than the master and standby are called members. These devices do not have any network, control, or management plane functions. Their interfaces are directly controlled and managed by the master switch.

Is there any restriction in the order of VSF member numbering?

There is no restriction on the order in which VSF members can be numbered. Each member, however, must have a unique number in the range of 1-10 (for 6300 switches) or 1-8 (for 6200F switches).

What is the supported stack height and topology?

- 6200F platforms can stack up to 8 members with no modular SKU (only fixed SKU).
- 6300 F/M platforms can stack up to 10 members in a chain or ring topology.

Ring is the recommended topology. This topology requires that each member is configured with two VSF links, interconnecting each member with two other members in the stack.

On a two-member stack, configure only one VSF link that connects to its peer.

There is no concept of a ring topology in a two-member stack.

Can features be configured on a VSF link?

Once an interface becomes part of a VSF link, no standard network layer protocol or feature can run on that interface because it is part of the VSF stack fabric.

Will configurations in an individual member switch be retained after joining a stack?

Individual member device configurations are not retained after the switch is renumbered and becomes part of a stack.

How do the consoles of each member in a stack work?

The console of the master switch provides a full CLI that can be used to manage the stack. Consoles of other stack members, including the standby, have a limited set of CLI commands that are useful for troubleshooting the device from a stacking functionality standpoint.

How does an image upgrade for a stack work?

To upgrade a stack to a new firmware image, use the `copy <TFTP/SFTP> image` command to download the image to the device. The image will be downloaded to the stack master first and then be distributed to the other members of the stack automatically.

After downloading the firmware, reboot the stack using the `boot system <PRIMARY/SECONDARY>` command. This action completes the upgrade process.

Adding or rebooting individual members before the upgrade process is complete can cause the individual member to fail while joining the stack.

Can I add a member to the VSF stack when the member is running an image with a different version than the stack?

When a device joins a stack and its firmware version is different from the version on the master, the master will push its firmware copy to the device. Once the device receives a copy of the firmware, it will reboot and rejoin the stack, now running the same version as the master.

What happens when the VSF master switch goes down?

The standby switch, if present, will take the role of the master. The old master switch will boot and join the stack as the standby switch. This transition will be seamless with limited network impact.

In the absence of a standby (no secondary member configuration), master device failure causes the remaining VSF members to reboot and come back up. At this point, members will enter a state in which they are waiting for the master to come back up. During this time, front plane ports of the members will be down.

How do I recover a device that has not joined a stack due to misconfiguration?

The `vsf renumber-to` command is used to trigger a device to take up its new member number and light up its VSF links. This command causes the device to reboot, come back up and wait for messages from the stack master. If the VSF link is configured incorrectly or the member number is wrong, the device could be waiting in this state indefinitely.

To recover a device in this state, execute the following commands:

1. Execute the `ctrl+c` command on the device console. This action launches the recovery console.
2. Execute the `vsf-factory-reset` command on the recovery console.

This action resets the device to factory-default.

- The device will come back up as member ID 1 with no VSF link configuration.
- The device can be configured with the correct member number and VSF links.
- The `vsf renumber-to` command will trigger this configuration to take effect.

The recovery console also has commands that allow the user to copy support files to an external server. This functionality is useful for troubleshooting stacking-related issues.

How do the management ports of each member in a stack work?

In a stack, only the master management interface is active. The management interface can be assigned an IP address for device management purposes. When a master device fails, the standby becomes master and activates its management interface.

How does replacing the master switch in a stack work?

The replacement device must be of the same part number as the switch being replaced. You must also have a standby switch configured for replacing the master of a stack without losing configuration.

Complete the following steps:

1. Execute the `vsf switchover` command to trigger the standby switch to take over the master role.
2. Once the stack is up with the new master, remove all physical connections from the old master switch that is being replaced.
3. Configure VSF interfaces/links on the new device. It is of critical importance to match the interfaces configured on the switch being replaced.
4. Physically connect the new device to the stack through configured VSF links.
5. The new switch will join the stack and take up the role of standby.

What is the workflow for replacing a standby or member switch?

The replacement device must be of the same part number as the switch being replaced.

Complete the following steps:

1. Configure VSF interfaces/links on the new device. It is of critical importance to match the interfaces configured on the switch being replaced.
2. Renumber the new device to match the switch being replaced.
3. Physically connect the new device to the stack through configured VSF links.
4. The new switch will join the stack and take up the standby or a member role based on the secondary configuration for the stack.

What happens if a VSF link fails?

- If the stack topology is a ring, it will degenerate to a chain when a VSF link in the stack fails.
- If the topology is a chain, a VSF link failure will result in a stack being split into two independent stack fragments.
- When a stack splits and the master and standby of the stack become part of two different fragments, the standby takes up the master role for its fragment. Network disruption can result because the two fragments are simultaneously active. Aruba highly recommends enabling VSF split-detection to gracefully handle split brain scenarios.
- If a stack splits and the master and standby are in the same fragment with the other members on a different fragment, the members-only fragment will:
 - Reboot.
 - Come back up.
 - Wait for communication from the stack master.

What is VSF split-detect?

When a stack splits, the split-detect feature provides a mechanism for the fragments to discover each other.

Once the two stack fragments are discovered, the fragment that has the primary member becomes the active fragment and keeps its front plane (non-VSF) interfaces up and running. The other fragment becomes inactive and all non-VSF interfaces on the inactive fragment are brought down to avoid network disruption.

How do I configure split-detect?

VSF supports split-detection through the management interface.

Connect the management interfaces of the primary and secondary members to the same management VLAN/network or connect them directly to one another. The CLI command to enable split detection is `vsf split-detect mgmt`.

How do I remove the non-VSF configurations in a stack?

Use the `erase startup-config` command on the VSF stack. This action will remove all non-VSF related configurations from the startup-config. Then reboot the stack.

Can a VSF member be removed from a stack?

Yes, remove a member from the stack using the `no vsf member <MEMBER-ID>` command. All configurations associated with the member will also be removed. The member will boot and come back up with the factory default configuration.

How do I remove the master switch from the stack?

Aruba does not recommend removing a member that is master of a stack.

If the master switch has to be removed, complete a switchover and wait for:

- the standby to take up the master role, and
- the old master to reboot and join the stack as standby.

Then use a member remove command to remove the device from the stack.

How can I boot the whole VSF stack and individual members using CLI?

The `boot system` command can be used to boot the whole stack.

To boot an individual member, use the `vsf member <MEMBER-ID> reboot` command.

Is modifying the VSF-specific configuration using Checkpoint restore or TFTP/SFTP/USB download supported?

This functionality is not supported. Before applying a configuration on a stack through Checkpoint restore or TFTP/SFTP/USB download, you must ensure that the following configurations match exactly:

- The current stack VSF configurations.
- The VSF configurations that are part of the configuration file that is being restored or downloaded from the server.

Specifically, the current VSF stack and the Checkpoint/downloaded configuration that will be applied on the stack must have the same:

- Number of members
- Member part number (J#)
- Member number
- VSF link configurations
- Secondary member configuration
- Split-detect configuration

How can I dismantle a stack?

A VSF stack can be dismantled by using the `erase all zeroize` command.

This action will cause each member to reboot, come back up with factory defaults, and function as individual/standalone devices.

How do I collect support files for a stacked device?

The `copy support-files all` command executed on the master console will collect support and troubleshooting information from all members that are part of the stack.

If a member is not part of the stack, you must run the same command from the recovery console of the respective member.

If a stack has split into two fragments, both fragments will have a master. Execute the same command on the master console of both fragments.

Why do I see the message 'Warning - management/VSF role decision pending, HPE Credential Manager daemon may shut down and restart.'

During the boot time, this message might appear due to a delay in discovering the role, or if the VSF links took longer than usual to become operational. The message is only informational and will not cause any functionality impact.

Accessing Aruba Support

Aruba Support Services	https://www.arubanetworks.com/support-services/
Aruba Support Portal	https://asp.arubanetworks.com/
North America telephone	1-800-943-4526 (US & Canada Toll-Free Number) +1-408-754-1200 (Primary - Toll Number) +1-650-385-6582 (Backup - Toll Number - Use only when all other numbers are not working)
International telephone	https://www.arubanetworks.com/support-services/contact-support/

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
Software licensing	https://lms.arubanetworks.com/
End-of-Life information	https://www.arubanetworks.com/support-services/end-of-life/
Aruba software and documentation	https://asp.arubanetworks.com/downloads

Accessing updates

You can access updates from the Aruba Support Portal or the HPE My Networking Website.

Aruba Support Portal

<https://asp.arubanetworks.com/downloads>

If you are unable to find your product in the Aruba Support Portal, you may need to search My Networking, where older networking products can be found:

My Networking

<https://www.hpe.com/networking/support>

To view and update your entitlements, and to link your contracts and warranties with your profile, go to the Hewlett Packard Enterprise Support Center **More Information on Access to Support Materials** page:

<https://support.hpe.com/portal/site/hpsc/aae/home/>

Access to some updates might require product entitlement when accessed through the Hewlett Packard Enterprise Support Center. You must have an HP Passport set up with relevant entitlements.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

<https://asp.arubanetworks.com/notifications/subscriptions> (requires an active Aruba Support Portal (ASP) account to manage subscriptions). Security notices are viewable without an ASP account.

Warranty information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

Regulatory information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at <https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

Additional regulatory information

Aruba is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see <https://www.arubanetworks.com/company/about-us/environmental-citizenship/>.

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